Quantification: 877_wt_z073c1-2
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	10
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 10)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3052.5	2716	1789.94	px^2
major_axis_length_px	lacuna shape	98.8449	103.026	32.5447	px
minor_axis_length_px	lacuna shape	40.3072	40.5486	15.7541	px
aspect_ratio	lacuna shape	2.628	2.2321	1.0659	unitless
eccentricity	lacuna shape	0.8894	0.8935	0.0708	unitless
solidity	lacuna shape	0.8013	0.8097	0.0523	unitless
roots_count	headline	11.3	9	7.5137	count
ring_length_r30_px	headline	413.3	356.5	217.025	px
ring_length_r60_px	reported	979.5	941	438.222	px
owned_length_px	ownership-dependent	1934.47	1471.63	1763.75	px
edge_count	ownership-dependent	95.5	61	92.386	count
mean_edge_length_px	ownership-dependent	19.0075	20.1247	7.0075	px
perimeter_px	normalised	310.044	299.676	105.33	px
ring_area_r30_px2	normalised	10426.7	10232.5	2662.66	px^2
ring_area_r60_px2	normalised	25344.7	25657	5710.76	px^2
in_frame_fraction_r30	normalised	0.9859	1	0.0445	unitless
in_frame_fraction_r60	normalised	0.9634	1	0.0826	unitless
ring_density_r30	normalised	0.0377764	0.0381433	0.0132811	px^-1
ring_density_r60	normalised	0.0371898	0.0382881	0.0107124	px^-1
roots_per_100px_perimeter	normalised	3.2839	3.5209	1.5901	per 100 px
ring_attached_length_r30_px	network v2	362.1	317	227.25	px
ring_attached_length_r60_px	network v2	757.8	657	516.819	px
ring_length_w_r30_px	network v2	472.981	415.318	247.018	px
ring_length_w_r60_px	network v2	1126.4	1076.97	505.816	px
sholl_crossings_r10	network v2	9.6	8	4.7422	count
sholl_crossings_r20	network v2	12	11	6.4291	count
sholl_crossings_r30	network v2	15.2	16	6.2147	count
loop_count	loops, report only	9.3	3.5	12.3473	count
loop_edge_share	loops, report only	0.2496	0.2586	0.1886	share
ring_width_mean_r30_px	width	6.6457	6.7405	0.8747	px

Per field
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measure	value	units
analysed_area_px2	1.01694e+06	px^2
total_skeleton_length_px	31426	px
canalicular_length_density_per_px	0.0309025	px^-1
junction_count	666	count
junction_density_per_px2	0.00065491	px^-2
lacuna_count	12	count

lacunae_per_px2	1.18e-05	px^-2
skeleton_component_count	367	count
mean_component_length_px	85.6294	px
median_component_length_px	24	px
field_length_density_w_per_px	0.035439	px^-1
field_density_without_flagged_per_px	0.0311489	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	498	px
unattached_component_count	27	count
canalicular_length_density_attached_per_px	0.0309025	px^-1
field_length_density_with_unattached_per_px	0.0313922	px^-1
loop_count	117	count
loop_edge_share	0.2788	share
edge_count	1643	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4481	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	29970	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	954	58.9902	25.4792	2.3152	0.9019	0.6974	1.3025	49.6237	860.655	False
2	6236	133.392	66.6525	2.0013	0.8662	0.7565	0.7065	193.518	518.807	False
3	1740	108.483	22.6499	4.7895	0.978	0.7716	1.1221	244.43	785.516	False
4	4847	142.75	45.7551	3.1199	0.9472	0.8253	0.4966	381.669	319.923	False
5	2583	119.873	29.7486	4.0295	0.9687	0.7747	0.5429	394.401	41.6202	False
6	4698	104.479	62.6265	1.6683	0.8004	0.7983	0.887	422.75	620.834	False
7	2761	84.1795	44.2264	1.9034	0.8509	0.8577	1.3929	435.645	810.324	False
8	3441	99.3913	46.2527	2.1489	0.8851	0.8428	1.0959	709.919	822.946	False
9	2671	101.574	36.8708	2.7549	0.9318	0.8211	0.7949	796.647	571.938	False
10	594	35.3373	22.8099	1.5492	0.7638	0.8672	0.7727	907.293	697.131	False
11	740	61.1371	17.5677	3.4801	0.9578	0.8675	-1.5683	1016.29	147.323	True
12								1017.52	842.011	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	351	769	2021.38	80	25.2673	6.043
2	25	793	1730	5467.12	277	19.7369	7.436
3	7	352	989	2005.23	101	19.8537	7.5283
4	19	518	1283	2957.39	145	20.3958	7.6229
5	5	227	592	528.375	28	18.8705	5.7804
6	19	732	1544	4072.96	222	18.3466	7.324
7	9	361	893	753.24	34	22.1541	7.269

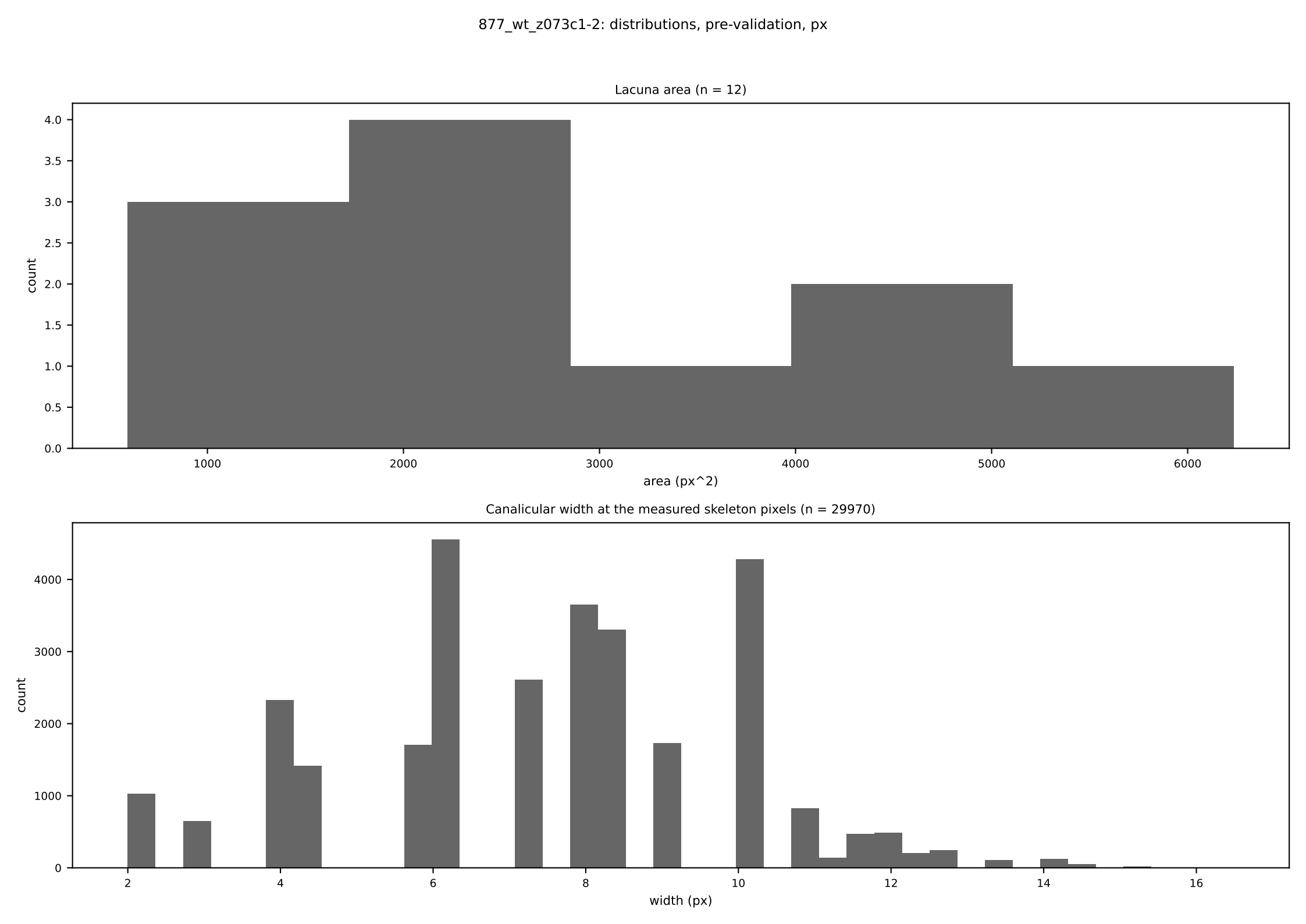
8	11	441	993	601.014	26	23.1159	6.212
9	9	255	718	938.034	42	22.3341	5.9929
10	0	103	284	0	0	0	5.2483
11	0	0	0	0	0	0	
12	0	6	103	0	0	0	3.2168

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	218.244	7625	18201	1	0.8799	0.0460328	0.0422504	4.1238
2	510.192	15054	34735	1	1	0.052677	0.0498057	4.9001
3	294.09	10047	25475	1	1	0.0350353	0.0388224	2.3802
4	387.789	13352	32069	1	1	0.0387957	0.0400075	4.8996
5	312.576	9613	20973	0.8593	0.7542	0.0236139	0.0282268	1.5996
6	397.546	12204	28960	1	1	0.0599803	0.0533149	4.7793
7	261.764	9629	24171	1	1	0.0374909	0.0369451	3.4382
8	305.262	10522	26302	1	1	0.0419122	0.0377538	3.6035
9	288.948	10418	25839	1	1	0.0244769	0.0277875	3.1147
10	124.024	5803	16722	1	1	0.0177494	0.0169836	0
11	159.918	3763	10281	0.5406	0.5269	0	0	0
12	93.598	3014	8836	0.5506	0.5329	0.00199071	0.0116569	0

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	310	538	406.09	897.477	8	8	16
2	768	1668	893.754	1984.24	15	23	19
3	324	895	415.889	1148.12	8	11	17
4	430	1163	594.571	1493.73	12	15	21
5	157	324	257.108	676.754	7	5	11
6	717	1377	841.311	1772.95	19	22	26
7	346	776	414.747	1019.55	10	11	13
8	307	484	503.274	1134.4	7	13	16
9	212	280	289.978	817.838	8	7	8
10	50	73	113.083	318.907	2	5	5
11	0	0	0	0	0	0	0
12	0	0	5.8284	118.054	0	0	1


877_wt_z073c1-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 29970)
count
width (px)